Sparks of Meaning #3: Consciousness

The Half-Life of Context: Rethinking Consciousness and AI

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What if consciousness isn’t mysterious at all and it’s just a matter of memory decay?

We humans live inside a rolling window of memory. We take in words, images, feelings: however, unless we reinforce them, they fade. Psychologists have measured this decay since the 19th century:

- **Ebbinghaus (1885):** charted the *forgetting curve* — memory strength drops off steeply, then levels out.
- **Peterson & Peterson (1959):** showed that short-term memory slips away in seconds if not rehearsed.
- **Atkinson–Shiffrin (1968):** proposed the classic model where fragile short-term stores only sometimes consolidate into long-term memory.
- **Miller (1956):** quantified the limits of working memory: about 7 ± 2 items.
  Ebbinghaus charted the “forgetting curve,”

This fading is not a failure but a fundamental feature. Without it, our minds would drown in the noise of everything we've ever seen or heard. The decay acts like a half-life: each moment of context loses strength over time, unless refreshed. The result is a kind of balance and fragility, but is also a source of stability.

And this, for me, may be the simplest and most practical way to define consciousness: not as a mystical flame or metaphysical essence, but as the half-life of context. Our consciousness lives in that context window. That window of time is our here and now.

What about machines?

Large Language Models (LLMs) like GPT-4o (OpenAI, 2025) or Claude 3.5 don't work this way. They don't "forget." Within a fixed window (say 128,000 tokens, about a novel's worth of text), everything is retained equally. There's no fading, no half-life until the window overflows. At that point, the oldest tokens are chopped off.

- Humans: context decays continuously (half-life).
- LLMs: context is flat until it truncates.

This explains a lot about how differently we think. Humans rarely spiral into runaway loops because our decaying context dampens instability. LLMs, by contrast, sometimes collapse into repetition or drift, precisely because they lack this natural decay.

Transformers as dynamical systems

In my recent preprint (Pairwise Phase Space Embedding in Transformer Architectures, 2025), I argued that Transformers aren't just statistical engines but nonlinear dynamical systems. The attention mechanism reconstructs trajectories in a high-dimensional phase space, echoing Takens' theorem from chaos theory. Sentences are paths across manifolds, not just strings of probabilities.

This means that stability, whether in humans or machines, emerges from how those trajectories are regulated. For humans, the regulation is decay + reinforcement. For machines, it is truncation + static weights.

Why the half-life metaphor matters

What may be useful about this framing is its simplicity:

- It ties psychology (Ebbinghaus's forgetting curve, STM decay) to engineering (context windows in LLMs).
- It avoids mystical definitions of consciousness.
- It provides a testable hypothesis: if we simulate half-life in LLMs—introducing exponential decay into the attention window—will stability improve?

Imagine weighting (w) context tokens with a simple exponential function:

where T is the “memory constant.” Newer context is stronger, older context decays unless refreshed. That’s essentially how our brains already work.

Comparative snapshot

Here’s how the pieces line up:

Concept	Humans (Biology)	LLMs (Artificial)	Implication
Memory retention	Exponential decay (half-life)	Flat, no decay until truncation	LLMs risk instability
Reinforcement	Neural rehearsal, plasticity, oscillations	Static weights, no natural reinforcement	Needs synthetic refresh
Stability mechanism	Half-life damping	Hard cutoff	Fragility in long contexts
Consciousness definition	Experience of decaying/refreshing context	Not applicable—flat window	Suggests path to “machine consciousness”

Closing thoughts

Of course defining consciousness as the half-life of context is not a final word. It’s a starting point a useful fiction in Bertrand Russell’s sense i.e. it gives us a way to compare biological and artificial cognition without resorting to mysticism.

In us humans, decay stabilizes. In LLMs, truncation destabilizes. If we want machines that think more like us, perhaps the answer isn’t just longer context windows, but smarter ones: windows with half-lives. Importantly, that’s an experiment we can run and it might bring us closer to understanding not only artificial intelligence, but consciousness itself.

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